

Name: _____ Teacher: _____

YEAR 11 MATHEMATICS ASSESSMENT TASK – JUNE 2008

Time Allowed: 60 minutes

Full working should be shown in every question..

Marks may be deducted for careless or badly arranged work.

No liquid paper is to be used.

If a correction is to be made, one line is to be ruled through the incorrect answer.

Marks

1. Find the exact value of

a) $\tan 300^\circ$

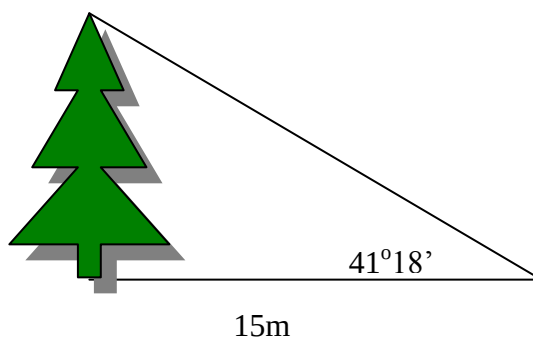
1

b) $\operatorname{cosec} (-225^\circ)$

1

2.

2



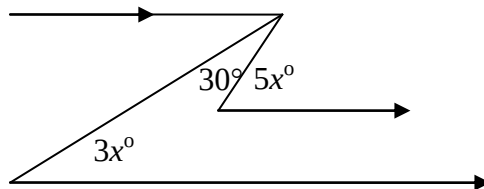
A tree casts a 15m shadow when the elevation of the sun is $41^\circ 18'$.
Find the height of the tree.

3. The interior angles of a regular polygon are 150°
How many sides does it have?

2

4. Find x (no reasons required)

1



5. If $\sin \theta = \frac{4}{7}$ and $\cos \theta < 0$ find the exact value of $\tan \theta$

2

6. If $\sin (2x + 20)^\circ = \cos (3x - 80)^\circ$ find x

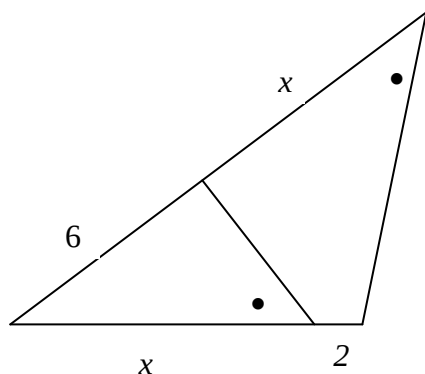
2

7. Show that $3x + 4y + 25 = 0$ is a tangent to the circle $x^2 + y^2 = 25$

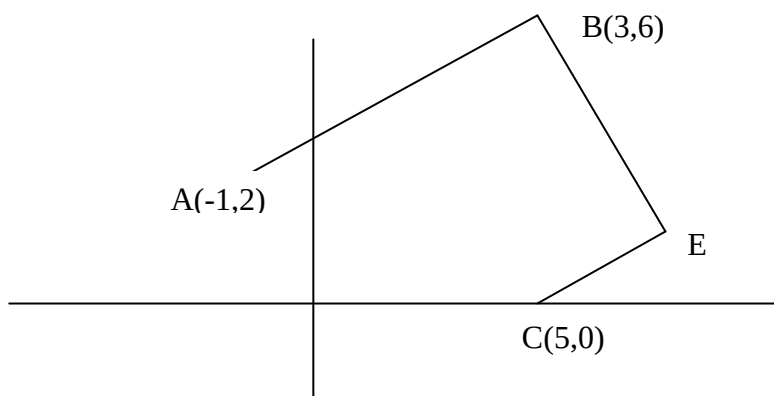
3

- | | | |
|----|--|--------------|
| 8. | Solve for $0 \leq \theta \leq 360^\circ$ | Marks |
| a) | $\sin \theta = \frac{\sqrt{3}}{2}$ | 2 |
| b) | $2 \cos^2 \theta = 2 + \sin \theta$ | 3 |
| c) | $\sec 2\theta = \cos 2\theta$ | 3 |

9. Find the exact value of x in the following, if the triangles below are similar. 4



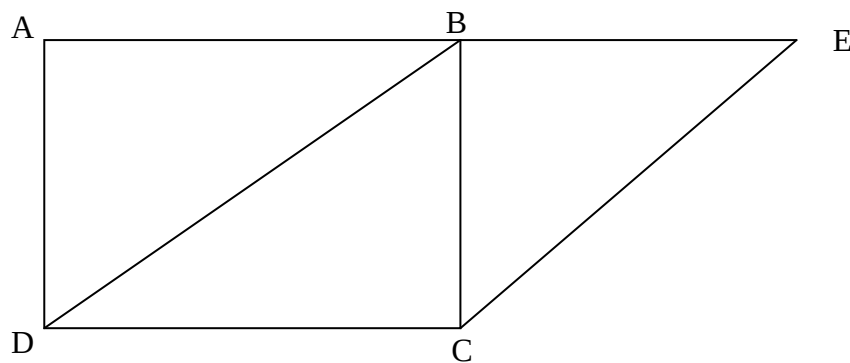
10.



- | | | |
|------|---|---|
| i) | Find the gradient of AB | 1 |
| ii) | Find M, the midpoint of AB | 1 |
| iii) | Find the equation of the line ℓ through M perpendicular to AB | 2 |
| iv) | Show that C(5,0) lies on the line ℓ | 1 |
| v) | If AB has equation $x - y + 3 = 0$ find the perpendicular distance from C to AB | 2 |
| vi) | If MCEB is a rectangle find the area of the trapezium ABEC | 3 |

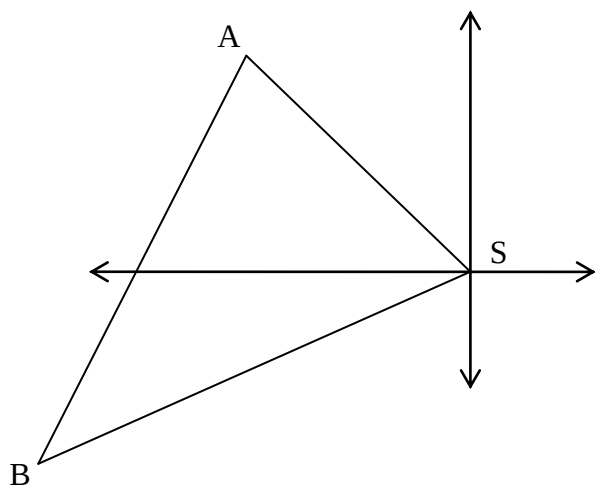
11.

Marks



ABCD is a rectangle. AB is produced to E such that $BD \parallel CE$

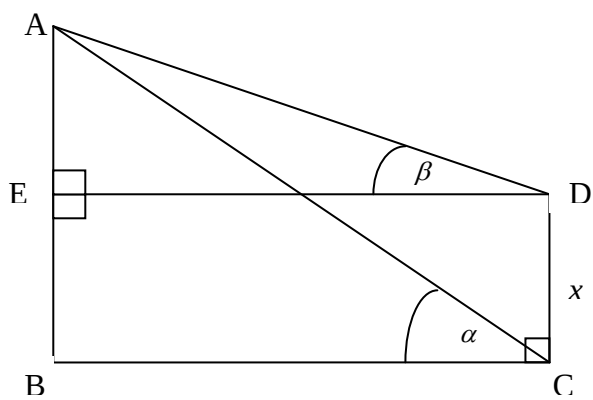
- i) Prove $\triangle ABD \equiv \triangle BCE$ 3
 - ii) If $AB : BC = 2 : 1$ find $\angle BEC$ 2
12. a) Sketch the graphs of $y = \cos x$ for $0 \leq x \leq 360^\circ$ and $y = \frac{1}{2}$ on the same set of axes 2
- b) For what values of x in the domain $0 \leq x \leq 360^\circ$ is $\cos x \geq \frac{1}{2}$. 2
13. Two planes leave Sydney at the same time. One flies 300nm northwest to Point A. The other flies 420nm on a bearing of 251° to Point B.



- i) Show $\angle ASB = 64^\circ$ 1
- ii) What is the distance AB? 2
- iii) What is the bearing of B from A? 3

14. Show $\frac{\cot \theta - \tan \theta}{\cos \theta - \sin \theta} = \operatorname{cosec} \theta + \sec \theta$

15.



i) Show $AD = \frac{BC}{\cos \beta}$

2

ii) Prove $BC = \frac{x \cos \alpha \cos \beta}{\sin(\alpha - \beta)}$

3

END OF EXAMINATION

$$1. a) \tan 300 = -\tan 60^\circ$$

$$= -\sqrt{3}$$

$$b) \operatorname{cosec}(-225)$$

$$= \frac{1}{\sin 135^\circ}$$

$$= \frac{1}{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{2}$$

$$2. \tan 41^\circ 18' = \frac{x}{15}$$

$$x = 15 \tan 41^\circ 18'$$

$$= 13.18$$

$$3. \frac{180(n-2)}{n} = 150^\circ$$

$$180n - 360 = 150n$$

$$30n = 360$$

$$n = 12$$

$$\therefore 12 \text{ sides.}$$

$$4. 3x + 30 = 5x$$

$$x = 15^\circ$$

Solutions:-

$$5. \begin{array}{c} 7 \\ \backslash \\ \theta \\ / \\ \sqrt{33} \end{array} + \tan \theta = -\frac{4}{\sqrt{33}}$$

$\sqrt{33} \rightarrow$ by Pyth. 2

$$6. 2x + 20 + 3x - 80 = 90^\circ$$

$$5x - 60 = 90$$

$$x = 30^\circ$$

$$7. \begin{array}{c} \swarrow \\ 3x + 4y + 25 = 0 \end{array} \text{ (for algebraic solution see back page)}$$

If $3x + 4y + 25 = 0$ is a tangent to $x^2 + y^2 = 25$

Then $d = 5$ units.

$$\therefore \frac{|3(0) + 4(0) + 25|}{\sqrt{3^2 + 4^2}} = d$$

$$d = \frac{25}{5}$$

$$d = 5$$

$$\therefore 3x + 4y + 25 = 0 \text{ is a tangent}$$

$$8. a) \sin \theta = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = 60^\circ, 120^\circ$$

$$b) 2\cos^2 \theta = 2 + \sin \theta$$

$$2(1 - \sin^2 \theta) = 2 + \sin \theta$$

$$\therefore 2\sin^2 \theta + \sin \theta = 0$$

$$\sin \theta (2\sin \theta + 1) = 0$$

$$\sin \theta = 0 \quad \sin \theta = -\frac{1}{2}$$

$$\therefore \theta = 0, 180^\circ, 360^\circ, 210^\circ, 330^\circ$$

$$9. 8c) \text{ See bottom of last page}$$

$$\frac{x}{x+6} = \frac{6}{x+2}$$

$$x(x+2) = 6x+36$$

$$x^2 + 2x - 6x - 36 = 0$$

$$x^2 - 4x - 36 = 0$$

$$x = \frac{4 \pm \sqrt{16 - 4(1)(-36)}}{2}$$

$$= \frac{4 \pm \sqrt{160}}{2}$$

$$= \frac{4 \pm 4\sqrt{10}}{2}$$

$$= 2 \pm 2\sqrt{10} \text{ but } x > 0$$

$$\therefore x = 2 + 2\sqrt{10}$$

$$(\text{accept } \frac{4 + 4\sqrt{10}}{2})$$

$$10. (i) m_{AB} = \frac{6-2}{3-1} = 1$$

$$(ii) M\left(\frac{3+1}{2}, \frac{6+2}{2}\right)$$

$$= M(1, 4)$$

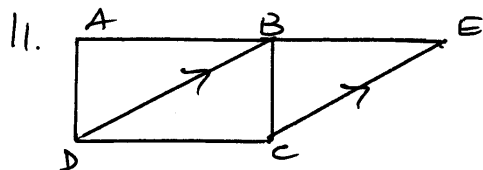
(iii) $y - 4 = -1(x - 1)$
 $y = -x + 5$ 2
 (iv) sub in (5)
 $0 = -5 + 5$
 $0 = 0$
 $\therefore (5, 0)$ lies on 1
 $y = -x + 5$

(v) $d = \frac{|1(5) + -1(0) + 3|}{\sqrt{1^2 + (-1)^2}}$
 $d = \frac{8}{\sqrt{2}}$ 2

(vi) Area of trapezium
 $= 3 \times \Delta AMC$
 $AM = \sqrt{(1 - -1)^2 + (4 - 2)^2}$
 $= \sqrt{8}$
 $\therefore \Delta AMC = \frac{1}{2} \cdot \sqrt{8} \cdot \frac{8}{\sqrt{2}}$
 $= \frac{1}{2} \cdot 2\sqrt{2} \cdot \frac{8}{\sqrt{2}}$
 $= 8$
 $\therefore \text{Area of Trapezium} = 24 \text{ unit}^2$ 3

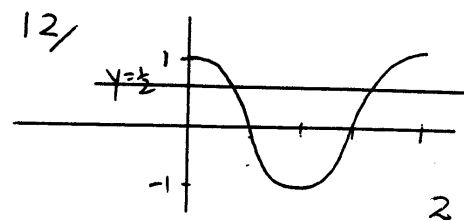
Alternative E(7, 2)

$AB = \sqrt{32}$ $CE = \sqrt{8}$
 $\therefore \text{Trop.} = \frac{1}{2} \cdot \frac{8}{\sqrt{2}} (\sqrt{32} + \sqrt{8})$
 $= \frac{8}{2\sqrt{2}} (4\sqrt{2} + 2\sqrt{2})$
 $= \frac{48\sqrt{2}}{2\sqrt{2}}$
 $= 24 \text{ units}^2$ (3)



(i) $\angle DAB = 90^\circ = \angle ABC$ (Angles of a rect.)
 $\angle CBE = 90^\circ$ (straight $\angle$)
 $\therefore \angle DAB = \angle CBE$
 $AD = BC$ (opposite sides of a rectangle)
 $\angle ABD = \angle AEC$ (corresponding $\angle$ s on $\parallel$ lines)
 $\therefore \Delta ABD \equiv \Delta BEC$ (AAS) 3

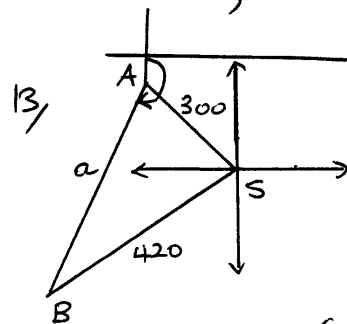
(ii) $\tan \angle BEC = \frac{1}{2}$
 $\therefore \angle BEC = 26^\circ 34'$ 2



$\cos x = \frac{1}{2}$ when $x = 60^\circ, 300^\circ$

$\cos x \geq \frac{1}{2}$ for

$0 \leq x \leq 60^\circ, 300^\circ \leq x \leq 360^\circ$ 2



(i) $\angle ASB = 360 - (251 + 45)$
 $= 64^\circ$ 1

(ii) $a^2 = 300^2 + 420^2 - 2(300)(420)\cos 64^\circ$
 $a = 394.88 \text{ n.m.}$ 2

$$(11) \frac{\sin \angle BAS}{420} = \frac{\sin 64}{394.88}$$

$$\sin \angle BAS = \frac{420 \sin 64}{394.88}$$

$$\angle BAS = 72^\circ 56'$$

$$\therefore \text{Bearing} = 116^\circ + 72^\circ 56'$$

$$= 188^\circ 56' 3$$

$$14, \frac{\cot \theta - \tan \theta}{\cos \theta - \sin \theta} = \frac{\cos \theta}{\sin \theta} + \sec \theta$$

LHS =

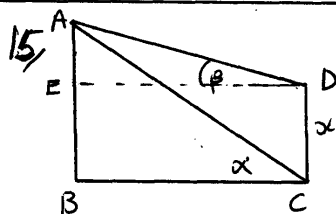
$$\frac{\cot \theta - \tan \theta}{\cos \theta - \sin \theta}$$

$$\frac{\cos \theta - \sin \theta}{\sin \theta \cos \theta} \div \frac{1}{\cos \theta - \sin \theta}$$

$$= \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{(\sin \theta \cos \theta)(\cos \theta - \sin \theta)}$$

$$= \frac{1 \cos \theta}{\sin \theta \cos \theta} + \frac{\sin \theta - 1}{\sin \theta \cos \theta}$$

$$= \sec \theta + \sec \theta 3$$



$$(i) \cos \beta = \frac{DE}{AD} \text{ but } DE = BC$$

$$\therefore \cos \beta = \frac{BC}{AD}$$

$$\therefore AD = \frac{BC}{\cos \beta} \quad \text{--- (1) } 2$$

$$(ii) \angle CAD = \alpha - \beta$$

$$\therefore \frac{x}{\sin(\alpha - \beta)} = \frac{AD}{\sin(90 - \alpha)}$$

$$\therefore AD = \frac{x \sin(90 - \alpha)}{\sin(\alpha - \beta)}$$

$$= \frac{x \cos \alpha}{\sin(\alpha - \beta)}$$

$$\text{from (1) } BC = AD \cos \beta$$

$$\therefore BC = \frac{x \cos \alpha \cos \beta}{\sin(\alpha - \beta)} \quad 3$$

Question 7. (Algebraically)

$$3x + 4y + 25 = 0 \Rightarrow y = \frac{-25 - 3x}{4}$$

$$x^2 + y^2 = 25$$

$$\therefore x^2 + \left(\frac{-25 - 3x}{4}\right)^2 = 25$$

$$x^2 + \frac{625 + 150x + 9x^2}{16} = 25$$

$$16x^2 + 625 + 150x + 9x^2 = 400$$

$$25x^2 - 150x + 225 = 0$$

$$x^2 - 6x + 9 = 0$$

$$(x - 3)^2 = 0$$

$$\therefore x = 3$$

$$y = 4 \quad (3, 4)$$

$\therefore$ Only 1 pt of intersection

$\therefore 3x + 4y + 25 = 0$ is a tangent to $x^2 + y^2 = 25$.

$$8c) \sec 2\theta = \cos 2\theta$$

$$\frac{1}{\cos 2\theta} = \cos 2\theta$$

$$\cos^2 2\theta = 1$$

$$\cos 2\theta = \pm 1 \quad 3$$

$$\therefore 2\theta = 0, 180, 360, 540, 720$$

$$\theta = 0, 90, 180, 270, 360$$